

Deep learning models for bridge deck evaluation using impact echo

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HIGHLIGHTS

- An annotated and validated impact echo dataset was formed.
- Deep learning was used to classify impact echo signals for the first time.
- Both 1D and 2D deep learning models were investigated.
- The proposed 1D model was superior to both conventional and other neural networks.

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ABSTRACT

Impact echo (IE) is a common nondestructive evaluation (NDE) method to detect subsurface defects in concrete bridge decks. The conventional approach for analyzing the IE data requires user expertise to define analysis parameters that could hinder broad field implementation. In this paper, the feasibility of using deep learning models (DLMs) for autonomous subsurface defect detection in bridge decks using IE has been investigated. A set of eight laboratory-made reinforced concrete bridge specimens with artificial defects were constructed at the Federal Highway Administration Advanced Sensing Technology NDE laboratory. A total number of 2016 of IE data was collected from these specimens. A one-dimensional (1D) convolutional neural network (CNN), and a 1D recurrent neural network using bidirectional long-short term memory units, were developed and applied on the IE data. In addition, two-dimensional (2D) world renowned CNNs were applied on the 2D representatives of the IE data, i.e., spectrograms. The proposed 1D CNN was the most accurate model achieving an overall accuracy of 0.88 by classifying 0.70 of the defects and 0.95 of the sound regions correctly. The proposed 1DCNN was superior to previous machine learning models that were previously used for IE classification. The results of this study showed the feasibility and the potentials of the DLMs for subsurface defect detection.

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1. Introduction

Bridge inspection has been included in the Code of Federal Regulations for more than half a century [1]. Conventional visual inspections [1–2] are time-consuming and dependent on the skill and experience of individual inspectors [3–6]. Considering only one type of infrastructure (bridges) and one bridge element (decks), inspectors must perform periodical inspections over almost 4 billion square meters of the US transportation grid [7], which is equivalent to an area larger than the City of Detroit in the United States. Bridge decks start exhibiting more defects

as they become older, making visual inspections even more burdensome.

Recent demand to introduce automation in infrastructure construction, inspection, and planning has led to implementing artificial intelligence (AI) for bridge evaluation. One can define a set of features in inspection data and employ AI to create statistical classifiers to detect bridge defects, i.e., conventional machine learning techniques. However, defining the features still requires an expert's opinion. Convolutional neural networks (CNNs) are a class of artificial neural networks, capable of obtaining both the features and the classifiers through training on annotated datasets to further reduce the role of experts in the classification process, i.e., deep learning.

Researchers have used machine AI for semi-autonomous surface defect detection from visual images to reduce the cost and time associated with manned practice, and to minimize the role

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of humans for interpreting the inspection data [8–10]. The application of AI for bridge condition evaluations has been mostly focused on the detection of surface defects in visual images [11–16]. Deep learning architecture has been shown to be superior to humans in terms of recognizing an object or a pattern in visual images [17]. Deep learning models (DLMs) also have been used in past studies to detect surface defects in bridges and other structures [18–21].

Bridge inspectors use nondestructive evaluation (NDE) techniques for comprehensive bridge deck evaluation as these methods provide essential information about the material properties and presence of subsurface defects. Examples of NDE techniques are sonic and ultrasonic methods [22,23], ground penetration radar [24], electrical resistivity [25], and impact echo (IE) [26–29]. A variety of AI techniques have been used for NDE data classifications [30,31]. Especially, researchers have developed classifiers and predication models to classify IE data [32–35]; however, these classifiers still required manual feature selection in the IE signals which limited their applications in the field. DLMs have not been designed for or implemented on IE data (except for bridge decks with overlay system [36]). DLMs are superior to statistical classifiers, i.e., unsupervised learning methods, because they depend on considerably fewer user-defined parameters [36,37].

The authors hypothesized that DLMs can be used for robust and unbiased IE data analysis as one of the essential steps towards introducing automation to bridge evaluation. The objectives of this paper include the followings.

- Forming a fully annotated dataset made of raw IE signals and their 2D transformation;
- Developing and training DLMs to classify defected and intact concrete;
- Identifying the most efficient DLMs for IE data analysis;
- Comparing DLMs to conventional method for IE data analysis;
- Validating DLMs on new IE data to mimic field inspections.

2. Background

2.1. Impact echo (IE)

IE instruments became commercially available in the late 1990s. American Society of Testing Material (ASTM) adopted this technology in 1998 to be formally used for thickness measurements of plate-shaped concrete structures [38]. A typical IE instrument includes an impact source, a nearby receiver, and a data acquisition system, as shown in Fig. 1. The impact source generates seismic waves on and through the inspected region. When the waves hit the bottom of a medium, i.e., backwall, or a subsurface irregularity, they are reflected. These reflections, i.e., echoes, then

travel back to the surface where their amplitudes will be recorded by receivers [36]. The travel time of the echoes could indicate the location of the reflectors, and whether a reflection is associated with a backwall reflection or a subsurface defect. Therefore, the IE test can be used to determine the thickness of the deck if applied on a sound region, or to detect subsurface defects if applied on a defected region [39].

2.2. The peak frequency method

The following relationship exists between the travel time of echoes in seconds (t), the echo's velocity in concrete in meter per second (C_p), and the traveled distance in meter (d):

$$t = \frac{d}{C_p} \quad (1)$$

If the echoes all correspond to the backwall reflection, then the travelled distance would roughly be equal to $2T$, where T is the thickness of the deck. Performing Fourier transform on IE signals, one can establish a relationship between the deck thickness and the frequency associated with it f_{IE} , as follows:

$$T = \beta \frac{C_p}{2f_{IE}} \quad (2)$$

where β is a correction factor influenced by the geometry of the propagation medium.

In theory, a distinctive peak in the frequency response (spectra), close to the thickness frequency (f_{IE}), indicates sound concrete. The presence of defects could be manifested in two distinct peaks, a single peak at higher frequencies than the f_{IE} , or a low-frequency response of the flexural mode [39]. However, these patterns do not always appear in the frequency response. Fig. 2a and b show the IE signals associated with a sound and a defected (shallow delamination) region on a concrete bridge specimen. The frequency responses exhibit a single peak, Fig. 2c and d. Without knowing the value of the thickness frequency or past engagement with the IE data, it is difficult to recognize the difference between the two. In addition, the shape of the frequency responses in these figures did not exactly follow the theoretical description of a defected concrete region.

Fig. 2e and f show the frequency responses after a Blackman Harris filter was applied. As seen, it is after the filtering that the frequency responses started to match with the theoretical interpretations. Note that the IE tests in Fig. 2 were performed in a controlled environment. IE signals and their frequency responses, collected in practice, tend to be noisier than the laboratory data which makes their classification even more difficult using the peak frequency method. Therefore, practitioners and inspectors demand

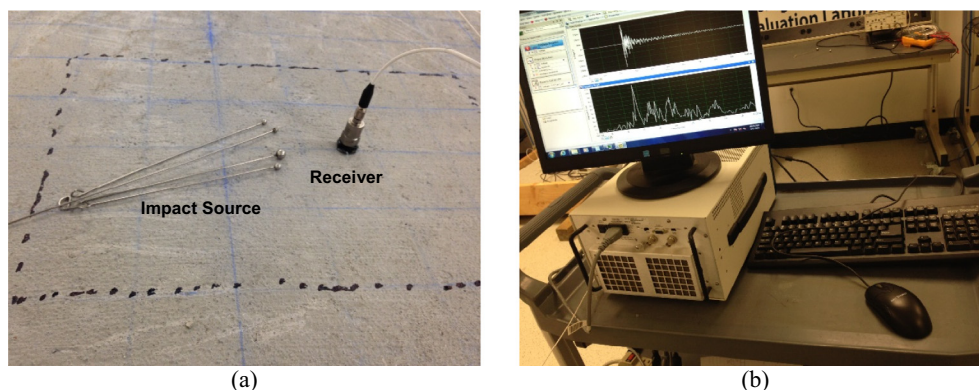


Fig. 1. (a) IE impact source and receiver, (b) data acquisition system.

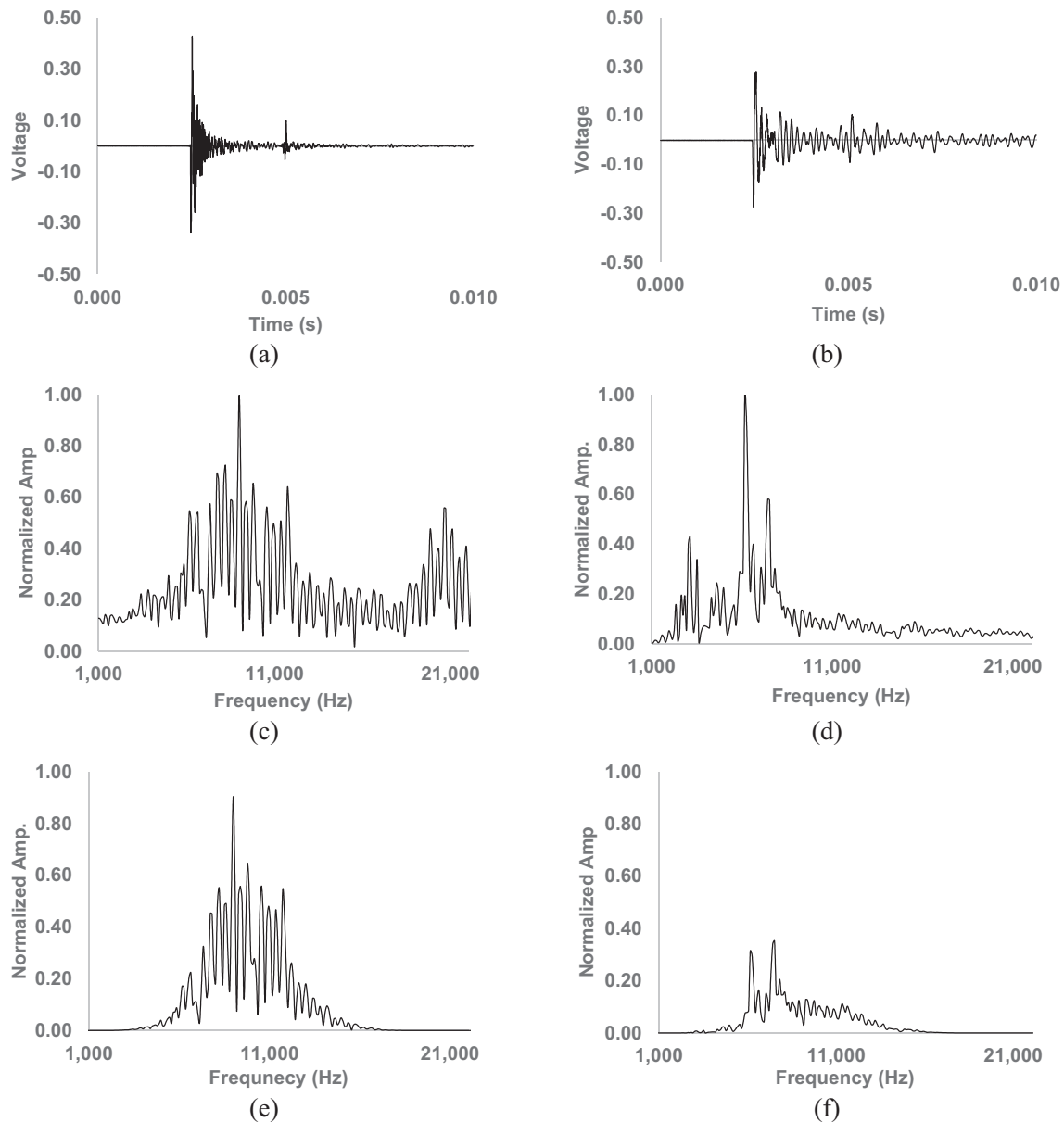


Fig. 2. IE signals (a) sound (b) defected (shallow delamination), frequency response without processing (c) sound, (d) defected, and frequency response with processing (e) sound, (f) defected concrete.

for alternatives that could produce consistent results. Unlike the peak frequency method that heavily relies on user-defined parameters and expertise, DLMs could potentially analyze raw IE data in an autonomous manner which could lead to a broader implementation of this technique in bridge inspection and evaluation.

3. Experimental program

3.1. Deck specimens

Eight reinforced concrete specimens, 3.0 m long, 1.0 m wide, and 0.2 m thick, were constructed at the Federal Highway Administration (FHWA) Advanced Sensing Technology (FAST) NDE laboratory with four types of artificial defects: shallow delamination (above the specimens' top rebar level), deep delamination (above the specimens' bottom rebar level), honeycombing, and voids, as seen in Fig. 3. These specimens were initially made to study the effectiveness of NDE methods on specimens with different types

of overlay systems [37]. The specimens were constructed by normal-weight concrete mix with a water-to-cement ratio of 0.37 and a 28-day compressive strength of 27.6 MPa, with two mats of uncoated steel reinforcement with no. 5 rebar (15.8 mm diameter) spaced at 203 mm in both directions. The specimens were inspected using several NDE methods including IE before and after placing the overlays. In this study, the IE data of specimens without overlays was investigated (for overlay specimens refer to reference [36]). All defects were 0.30 m long along the length of the specimens, and 0.20 m wide along the width of the specimens. A transverse crack was also artificially made in the middle of all specimens. More details about the defects, their materials, and specimen construction can be found in [37].

3.2. Annotated IE dataset

Each specimen was marked to create a surface grid system spacing 100 mm (longitudinally and transversely). The IE tests

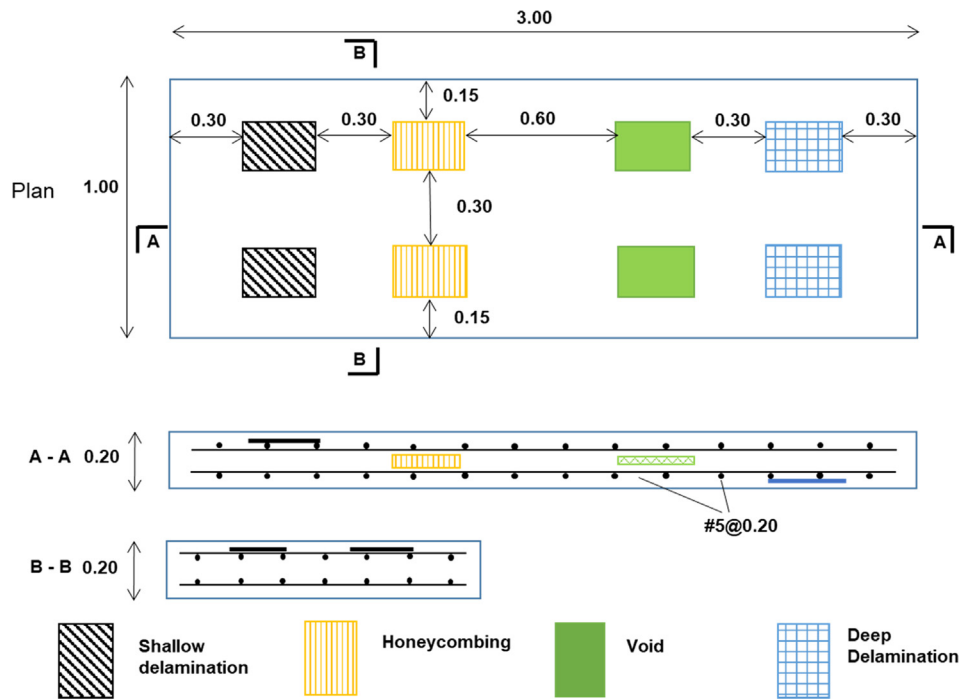


Fig. 3. Details of the specimens with artificial defects.

were performed on the grid points using a custom-made IE device, an accelerometer (receiver), and steel spheres (11 mm in diameter) on spring rods (impact source), as shown in Fig. 1. The sampling frequency was 200 kHz for 10 ms of IE data acquisition at any point of the girder. Therefore, each IE signal consisted of 2000 points (shown in Fig. 2a and b) and was saved by a name that represented the location of the receiver on the grid. The dataset can be accessed online [40]. No IE tests were conducted on and within 100 mm of the transverse crack.

Table 1 shows the classification of the IE test signals into sound and defected concrete, along with their quantities. In this table, class 0 was assigned to sound concrete and class 1 was assigned to the defected regions. Data collected on the grid points corresponding to boundaries of sound and defected regions were labeled as defected (class 1).

4. Investigated deep learning methods

Recurrent neural network (RNN) has been used for classification of time-variant signals. Hidden layer(s) in a RNN include the neurons that have a memory to preserve the information about the previous time step. In this study, a bidirectional long short-term memory (biLSTM) network was developed and implemented on the IE signals. A biLSTM network is a combination of two LSTM networks, one works on the signal in a normal order, the other works on the signal in a reverse order [41]. CNNs have been widely and successfully used for image classification in different disciplines, including defect detection in civil infrastructure. CNNs are commonly used on 2D input data, i.e., images; however, it is possible

to construct CNNs for 1D data as well [42,43]. In this study, a 1D CNN was developed to classify the IE signals.

To cover other possible analysis methods, 2D CNNs were investigated to classify 2D representatives of IE data. In this paper, short-time Fourier transform (STFT) was used to generate these representatives, i.e., spectrograms, as seen in Fig. 4. In the spectrograms, the horizontal axis represents time and the vertical axis represents frequency. The spectrograms provide the opportunity for implementing famous CNN architectures, commonly used in the realm of visual images, for IE classification. Note that the spectrograms were generated using all the existing frequencies (0–200 kHz), without removing the unwanted frequencies.

4.1. 1D deep learning

The biLSTM network architecture, used in this study, is shown in Fig. 5 (neurons in the middle layer are schematic). The network consisted of an input layer to read the IE data, a hidden layer (biLSTM layer with 200 neurons, a fully connected layer, and a soft-max layer), and an output (classification) layer to label the IE data.

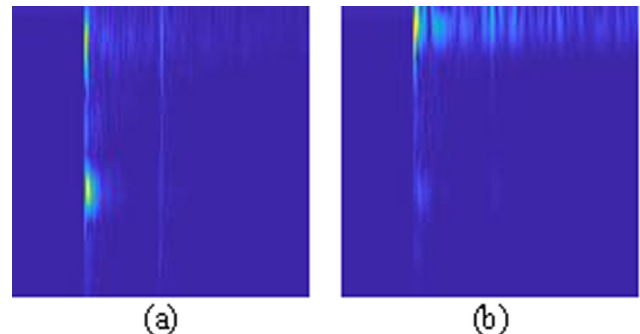


Fig. 4. Data 2D representatives of IE signals with the horizontal axis represents time and the vertical axis represents frequency: (a) sound concrete, (b) defected concrete (with shallow delamination).

Table 1
Classification and number of IE tests.

Type	Quantity	Class
Sound	1504	0
Defected	256	1
Boundary	256	1
Total	2016	

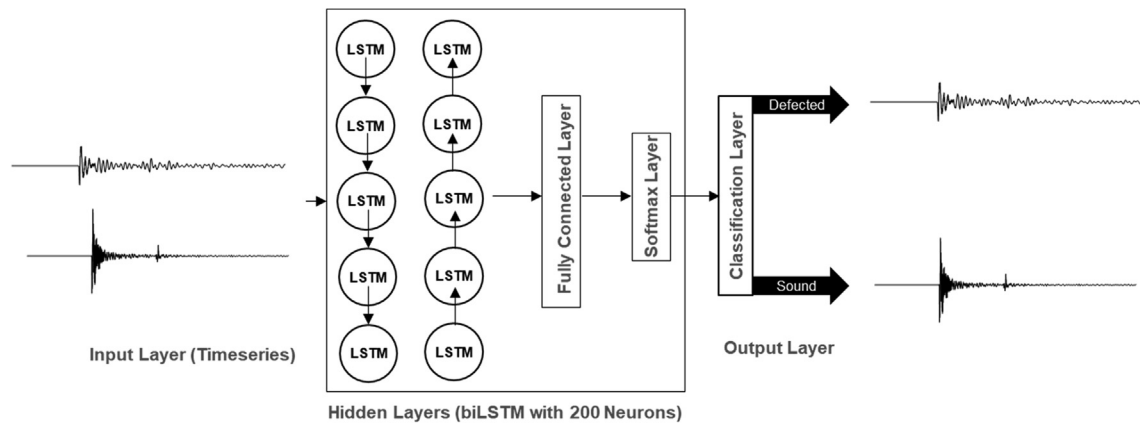


Fig. 5. Architecture of the biLSTM network.

The number of neurons was selected empirically and the network architecture was inspired by [41]. The fully connected layer connects the parameters from the past layers together in way that the softmax layer can assign a class probability to each signal. The classification layer assigns a label to the data, 0 or 1, according to the highest probabilities.

Fig. 6 shows the architecture of the proposed 1D CNN. This network is a combination conventional layers in deep learning literature including convolution, batch normalization, and rectified linear (CBR layers); max pooling (MP layers); fully connected, rectified linear unit, and dropout (FRD layers), and softmax layer. The size and number of convolution kernels for each CBR layer are shown Fig. 6. In CBR, a batch normalization is applied to activation maps which are the result of the convolution operation on the input from the previous layer. The batch normalization layer increases the speed of training by normalizing the activation maps. The rectified linear unit (ReLU) is an activation function that adds non-linearity to the normalized activation maps. The MP layer analyzes sets of neighboring locations in each activation map and represents them by their maximum value to reduce the size of the input and avoid overfitting. In the FRD units, dropout layers toss out responses smaller than specific threshold values.

4.2. 2D deep learning

Three world renown CNN architectures were trained and applied on the IE spectrograms: AlexNet [44], GoogleNet [45],

and ResNet [46]. Each has won the image classification competition (2012, 2014, and 2016, respectively) [47]. For brevity, the details of these networks are not discussed in detail here, given most of layers were already defined before. Fig. 7 shows the AlexNet schematic architecture.

Fig. 8 depicts a schematic architecture of GoogleNet. The significant contribution of this architecture to the realm of CNNs was the introduction and application of an “inception” layer (a combination of convolution and max pooling kernels). With an inception layer, a concatenation filter is applied to make the output of the inception layer compatible with the rest of network. AP in the GoogleNet architecture stands for average pooling and works similar to the MP layer, except it represents the previous layer activation maps with their average values instead of their maximum values.

Finally, the ResNet architecture is shown in Fig. 9. ResNet introduced residual connections (a combination of convolution and max pooling) between the convolution kernels to short-circuit the model for faster optimization.

4.3. Training and testing parameters

The annotated IE dataset was split into a training dataset and a testing dataset. The IE data of one specimen was used as the testing dataset, while the rest were used as the training dataset for each DLM. By designating IE data from different specimens as the testing dataset, eight training dataset were formed which then used

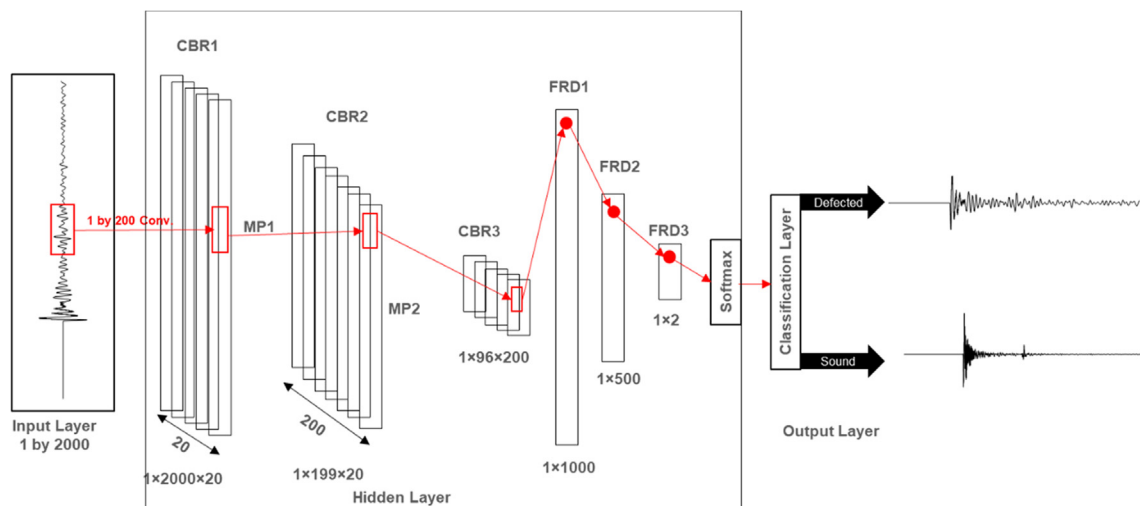


Fig. 6. Schematic. Architecture of the 1D CNN.

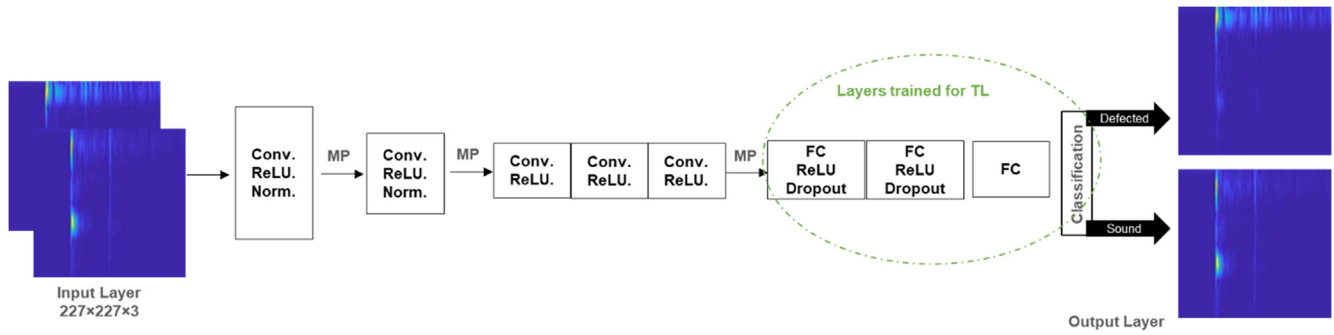


Fig. 7. AlexNet Schematic Architecture.

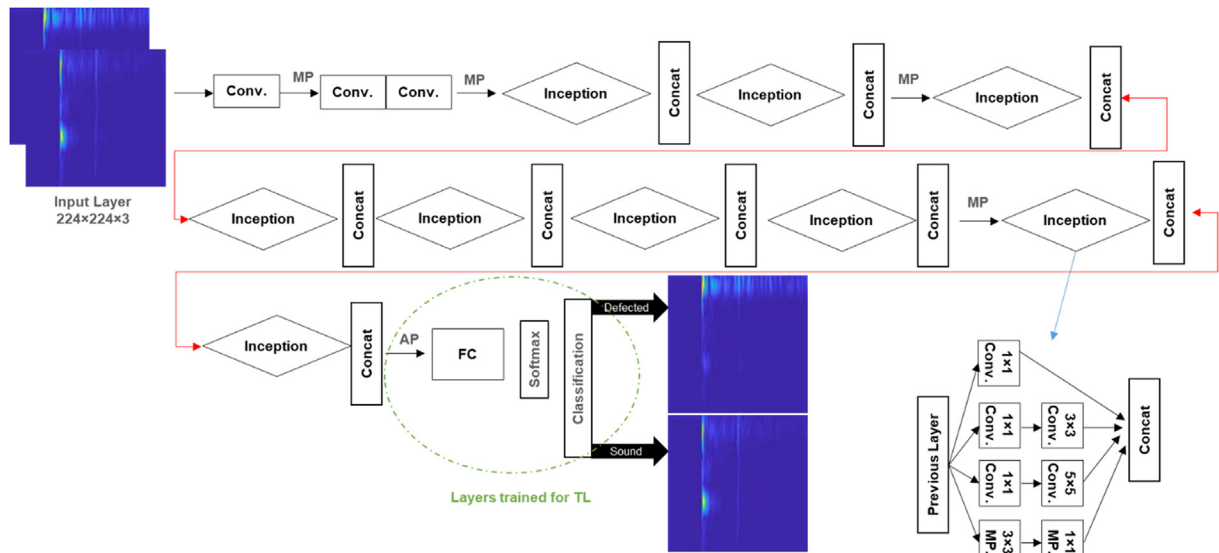


Fig. 8. GoogleNet schematic architecture.

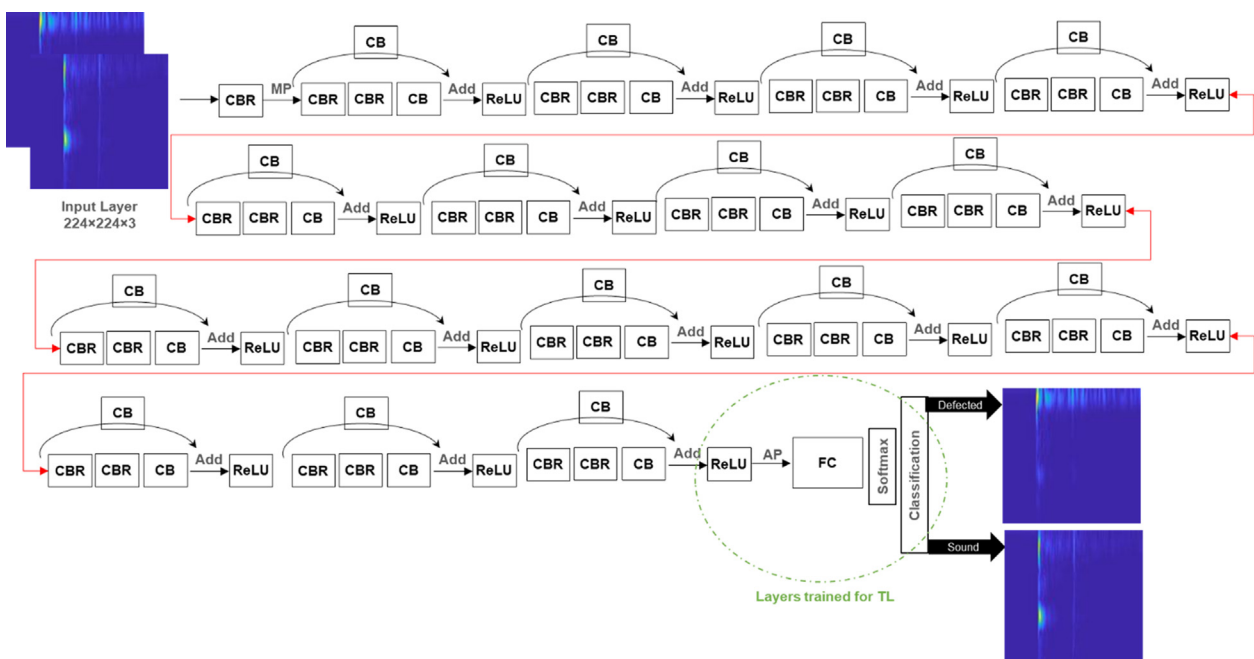


Fig. 9. ResNet schematic architecture.

Table 2
IE training and testing dataset.

Dataset	Class	Label	Quantity
Training	Sound	0	1764
	Defected	1	448
Testing	Sound	0	188
	Defected	1	64

to train the investigated DLMs. Table 2 shows the size and labels of the training and testing datasets.

The biLSTM and the 1D CNN were fully trained (FT) using the training datasets. For the 2D CNNs, a method called transfer learning (TL) was also used. In TL mode, parameters obtained from trained networks on large image datasets, i.e., ImageNet, are squeezed into the architecture of the new models before being trained on a new dataset, i.e., the IE data. Then, the weights in the end layers (with respect to the input layer) should be updated during the training. The weights in the first layers consist of generic feature extractors which potentially are useful for any classification task; while, the weights in the end layers are updated through re-training to match with the new dataset [30]. The TL technique is very effective when training deep architectures on small datasets. Therefore, AlexNet-based model was trained in both FT and TL manners while GoogleNet-based and ResNet-based models were only TL trained. The layers used for re-training in the TL mode for each DLM was different which are shown in Figs. 7–9 by dashed circles.

All computations were performed on a desktop with a 64-bit operating system, 32 GB memory, Intel® Core™ i7 CPU, and two GeForce GTX 1080 Ti graphics processing units (GPU). MATLAB 2019a was used to program the DLMs. Before training began, several training parameters, i.e., hyperparameters, were determined for each model. These parameters and their values (shown in Table 3) were chosen empirically to optimize the training process. In Table 3, solver refers to the type of algorithm used to optimize the model during the training. The purpose of an optimizer is to solve the gradient descent of the model in order to minimize the model's error. An adaptive moment estimation (ADAM) optimizer was selected for the biLSTM model while the stochastic gradient descent with momentum (SGDM) solver was chosen for the rest. Mini batch size is the size of the segments of the training data used to calculate the loss and error of the models. Iteration is when the models go through a forward and backward processing for a batch of data while one epoch is when all training data is processed once over the training dataset. Learning rate is the tolerance of parameters to optimize the gradient descent and minimize the loss. The number of retrained layers in each network is also shown in Table 3 (only applicable to transfer learning).

5. Results and discussion

The performance of a DLM was evaluated using binary classification metrics (equations (3) through (6)). The DLMs labeled an input signal or a spectrogram as the class that had the highest a

probability among the sound and defected classes. When a model labeled a defected input correctly, it was a true positive (TP) whereas false negative (FN) was when the model labeled it as sound. When the model labeled a sound input correctly, it was a true negative (TN) whereas false positive (FP) was when the model labeled it as defected. The true positive rate (TPR), true negative rate (TNR), accuracy (ACC), and F1 score were used to evaluate and compare the results of each model using the following equations:

$$TPR = \left(\frac{TP}{TP + FN} \right) \quad (3)$$

$$TNR = \left(\frac{TN}{TN + FP} \right) \quad (4)$$

$$ACC = \left(\frac{TP + TN}{TP + FN + TN + FP} \right) \quad (5)$$

$$F1 = \left(\frac{2TP}{2TP + FN + FP} \right) \quad (6)$$

5.1. Training results

The results of training on each specimen are shown in Fig. 10. The legends of these plots represent the specimen number that was used for testing. For instance, a training process labeled as S1 is when the DLMs are trained on specimens S2 through S8. All models reached an accuracy of 95 percent or higher at the last iteration of their training.

Each model required a different amount of time to finish one round of iteration based on its architectures. The 2D models were more time-consuming than the 1D models since more calculations were required to analyze the spectrograms compared to analyzing raw IE signal. As the models got deeper, they required more time to finish one iteration from AlexNet (TL) with 0.28 s to ResNet with 1.21 s. The AlexNet FT required 0.36 s per iteration since more weights had to be determined during training compared to TL. With 0.11 s per iteration, the biLSTM model required twice the time 1D CNN spent on one iteration, i.e., only 0.05 s, but 1D DLMs were both considerably faster than the 2D CNNs. As seen in Table 3, the biLSTM model required more iterations than convolutional networks to reach a desirable accuracy (e.g., 95%) at the end of the training. Taking this fact into consideration, the proposed 1D CNN was the fastest among investigated DLMs by a substantial margin.

5.2. Testing results

After the training was completed, the DLMs were used to classify the IE data in the testing dataset. The achieved performance metrics are shown in Fig. 11 (only for 1D CNNs), and Table 4 shows the mean, coefficient of variation (COV), maximum, and minimum values of the performance metrics (equations (3) through (6)) for each DLM from the eight testing results. The results are promising

Table 3
Training parameters for each investigated network.

Network	Mode	Solver	Mini Batch Size	Max Epochs	Learning Rate	Retrained Layers
biLSTM	FT	ADAM	20	95	1e-3	NA
1D CNN	FT	SGDM	50	40	1e-3	NA
AlexNet CNN	FT	SGDM	50	40	1e-3	NA
AlexNet CNN	TL	SGDM	50	40	1e-4	3
GoogleNet CNN	TL	SGDM	50	60	1e-4	4
ResNet CNN	TL	SGDM	50	25	1e-4	4

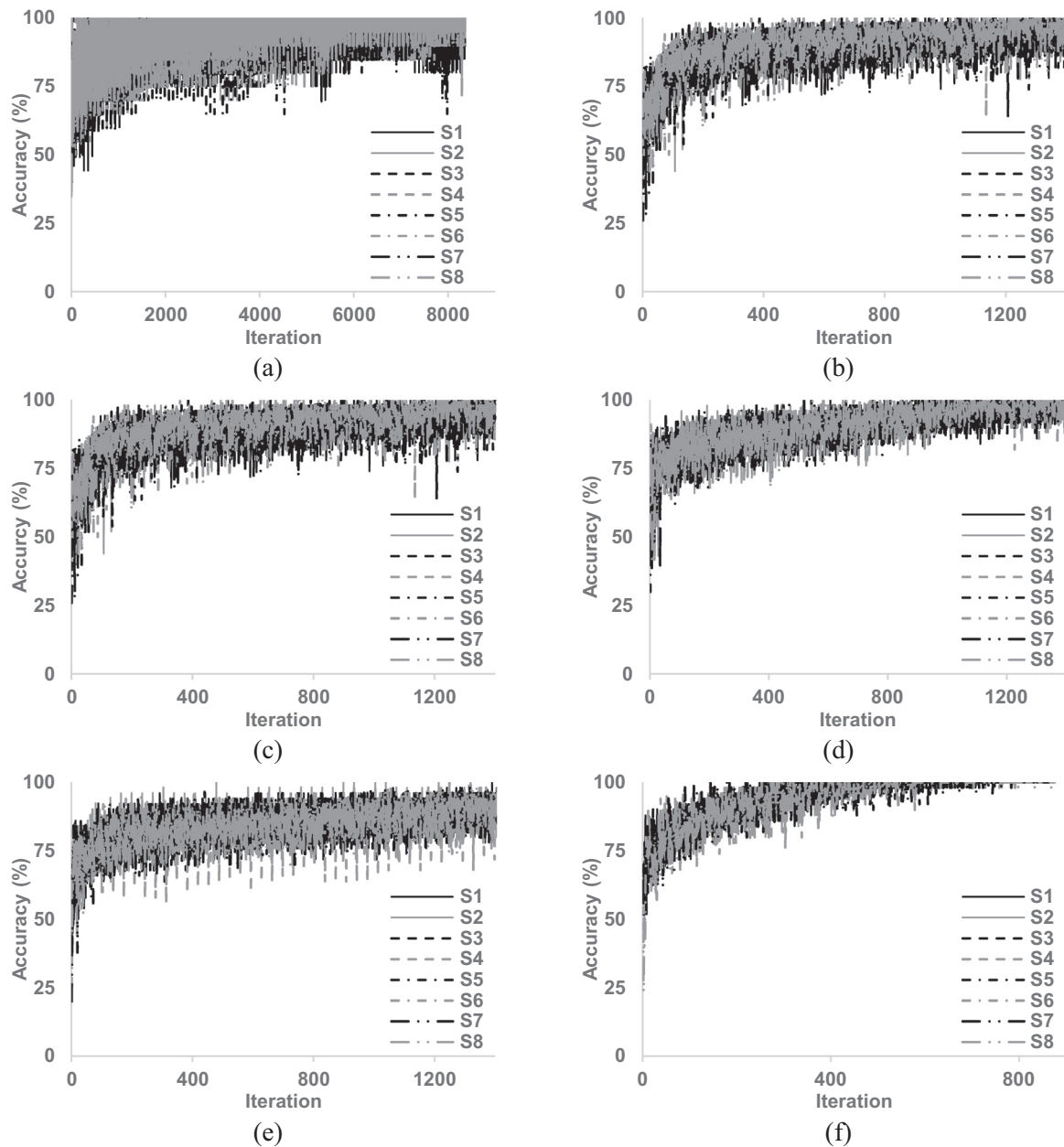


Fig. 10. Results of training using different deep learning models: (a) biLSTM, (b) 1D CNN, (c) AlexNet fully trained, (d) AlexNet (transfer learning), (e) GoogLeNet (transfer learning), (f) ResNet (transfer learning).

and relatively consistent (low COV values). The proposed 1D CNN produced the best average TPR, 0.70, with the lowest COV, 0.14. The ResNet model resulted in the lowest TPR values with only 0.53 on average. The 1D CNN model also produced the highest TNR and ACC on average with 0.95 and 0.88 respectively, which were consistent for all specimens (COV of 0.04). The biLSTM model produced the lowest average TNR and ACC values of 0.87 and 0.80, respectively. In terms of F1 scores, the 1D CNN was also the winner among the investigated models with 0.75; the ResNet and biLSTM models shared the low of 0.61. In addition, the 1D models were more accurate than the 2D models since they were applied directly on the signal, not the spectrograms. As seen the peak frequency method described in Section 2.2, the patterns exhibit themselves in a specific range of frequencies. The spectrograms on the other hand were constructed based on all possible frequencies. Unlike visual images, the vertical and horizontal axis in spectrograms rep-

resent time and frequency, which limited the 2D CNNs spatial invariance for feature detection. In addition, the pixels in the visual images often belong to an object whereas there are no objects in the spectrograms. The challenges of using DLMs on the spectrograms of signals have been previously observed in other studies [47–49]. Another noteworthy observation is the consistently higher than TPR compared to TNR. As shown in Table 2, the number of training data classified as 0, i.e., sound, was almost four times larger than the defected class which could justify this observation. However, unlike other applications of deep learning, this imbalance is benign since it matches the reality of in-service bridges. The second reason for lower true classification rate on the defected class was due to the variability of the defected regions compared to the sound concrete. The defects had four sources: shallow delamination, voids, honeycombing, and deep delamination, as shown in Fig. 3. This difference was likely manifested in

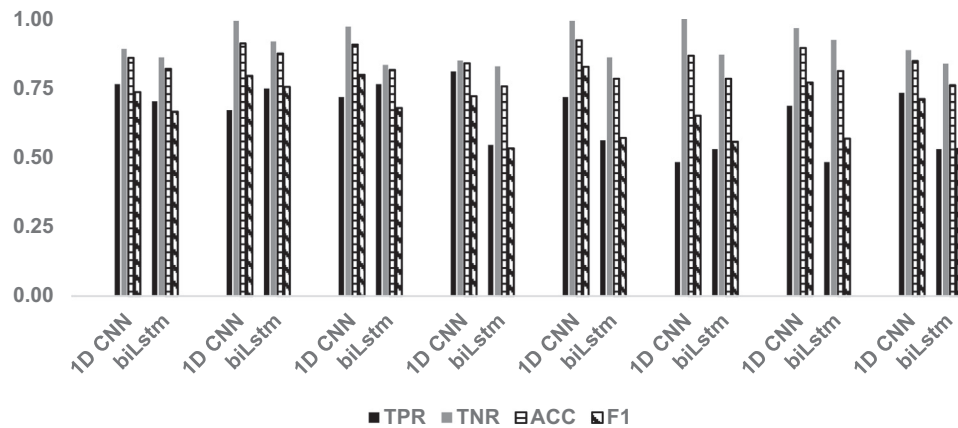


Fig. 11. Testing metrics for the investigated 1D DLMs using IE signals.

Table 4

Summary of performance metrics for the investigated DLMs.

Model	Metrics															
	TPR				TNR				ACC				F1			
	Mean	COV	Max	Min	Mean	COV	Max	Min	Mean	COV	Max	Min	Mean	COV	Max	Min
biLSTM	0.61	0.18	0.77	0.48	0.87	0.06	0.93	0.83	0.80	0.05	0.88	0.76	0.61	0.14	0.76	0.53
1D CNN	0.70	0.14	0.81	0.48	0.95	0.04	1.00	0.85	0.88	0.04	0.92	0.83	0.75	0.08	0.83	0.65
AlexNet (FT)	0.70	0.17	0.89	0.50	0.92	0.06	0.97	0.81	0.86	0.03	0.89	0.83	0.72	0.07	0.76	0.60
AlexNet (TL)	0.67	0.26	0.83	0.27	0.90	0.08	1.00	0.77	0.86	0.05	0.89	0.76	0.67	0.18	0.80	0.42
GoogleNet	0.64	0.19	0.83	0.41	0.91	0.05	0.98	0.87	0.84	0.02	0.87	0.81	0.66	0.09	0.75	0.55
ResNet	0.53	0.19	0.63	0.36	0.94	0.05	0.99	0.89	0.83	0.02	0.87	0.81	0.61	0.09	0.70	0.52

some of features extracted by the DLMs from the IE signals and consequently the spectrograms within the same class, i.e., 1. This issue can be mitigated by defining more classes for the dataset; however, each class would require more data than what was available in this research.

The performance of the 2D CNNs seems to be counterintuitive at first since the deeper networks resulted in poorer performance. The performance of these models does not always follow their performance on ImageNet [50–52]. In addition, deeper networks, e.g., GoogleNet and ResNet, do not necessarily perform better than shallower networks, e.g., AlexNet, for binary classifications [21]. For better implementation of 2D CNNs, more labeled spectrograms are required. Note that spectrograms could be generated differently, using other transforms such as wavelet, that could potentially alter the performance of the 2D CNNs.

To investigate the sources of false reports or the misclassified IE data are shown in Table 5. Almost all the mutual false negatives between the investigated DLMs were associated with the IE data collected on the boundary. Higher false negatives in some models (S6 compared to S3–S5) while models were trained on relatively similar datasets indicates that the observed inconsistency has external sources. It is possible that the IE instrument was not prop-

erly positioned during the data collection. In addition; defects may have been marginally misplaced on the boundaries; therefore, the associated IE signals were mislabeled and consequently misclassified.

5.3. Comparison

In this section the 1D CNN, as the best performing DLM, was selected to be compared to other methods of IE data classification.

5.3.1. Conventional method

Defect maps were used to compare the investigated DLMs to the peak frequency method. A defect map shows the relationship between the spatial location of subsurface defects to a certain property of the IE signal [35,48,52]. This property is the frequency associated with highest amplitude, e.g., vertical axis of the spectrums shown in Fig. 2. Using these frequencies, one can generate contour plots to present the defects locations. A similar approach was followed to create contour plots using the DLMs by plotting the probability assigned to the defected class which is essentially the probability of defect presence. Fig. 12 shows the defect maps generated by the peak frequency method and the best performing DLM, 1D CNN, for all specimens (S1 through S8).

The 1D CNN provided a precise prediction of defects locations for all four defect types in all specimens. In addition, it produced a better distinction between the sound and the defected concrete through a more tangible contrast change in the defect maps compared to the peak frequency method. Shallow delamination was detected easier by both approaches since IE technique is inherently better for shallower defects [37]. The proposed 1D CNN had difficulties when used on S6 and S8 as expected due to possible construction or data acquisition errors. The results show not only that deep learning is a feasible solution for IE data analysis but it

Table 5

Source of false negatives reports by DLMs.

Specimen	No FNs	Type (Quantity)
S1	8	Boundary (8)
S2	6	Boundary (6)
S3	2	Boundary (2)
S4	1	Boundary (1)
S5	3	Boundary (2)
S6	16	Boundary (13)
S7	8	Boundary (8)
S8	6	Boundary (6)

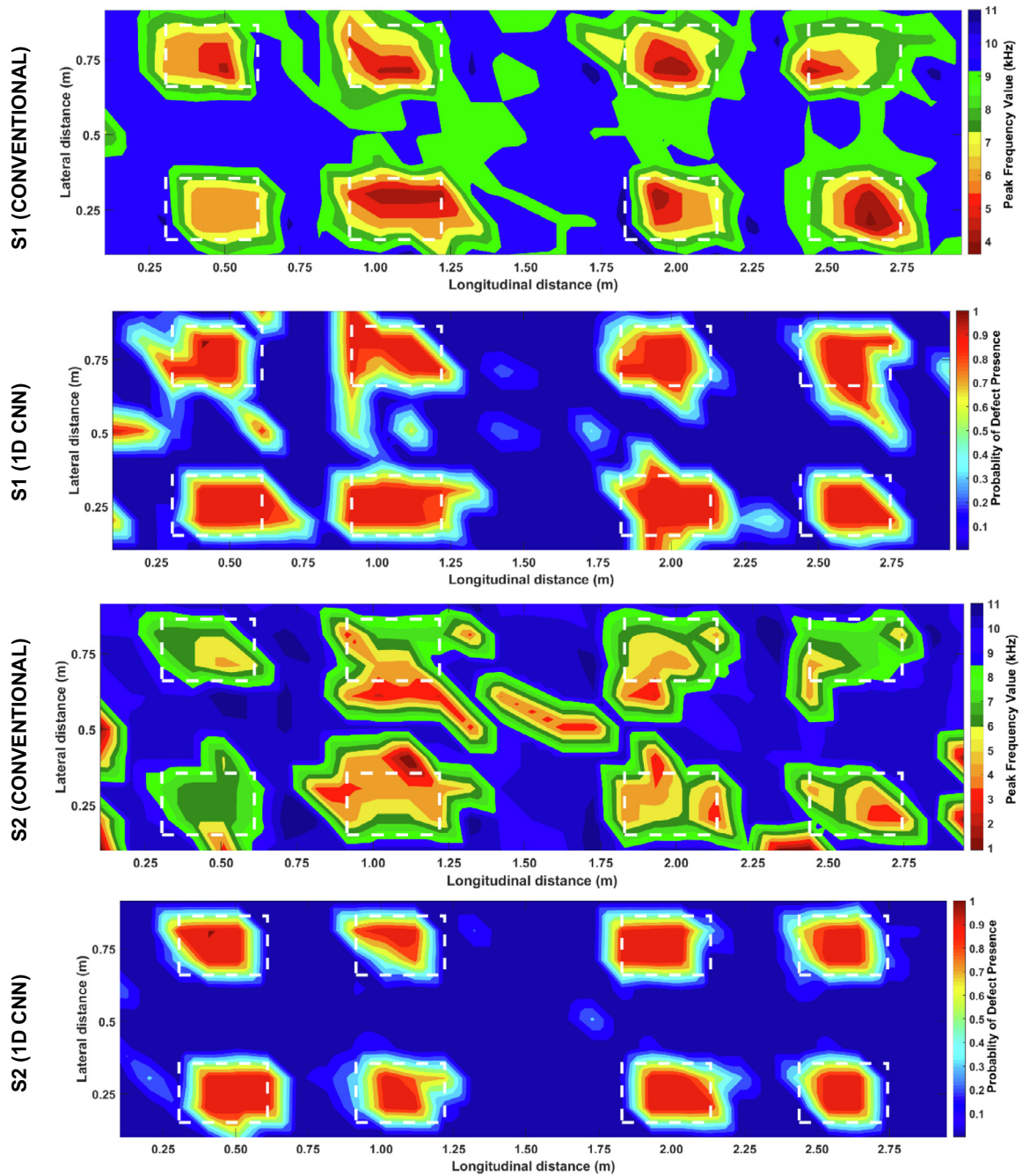


Fig. 12. Defect maps generated by frequency analysis and 1D CNN.

generated defect maps more robustly compared the conventional frequency analysis.

5.3.2. Comparison to machine learning models

IE classification has been done in the past using conventional machine learning techniques. In [33], classification features were captured from power spectrum density (PSD) functions of the IE data. The PSDs were estimated using fast Fourier transform of the original IE signals; and cumulative PSD area (CPA) and cumulative PSD ratio (CPR) were selected as features for classification (for more information refer to [33]). Another study extracted more statistical characteristics of the PSD (centroid, standard deviation, skewness, and kurtosis) along with the centralized signal energy as the features for IE classification [34]. These features were also

calculated for one wavelet approximation coefficient and two wavelet detail coefficients associated with each IE signal. The coefficients were the result of a four-level wavelet decomposition using second-level Daubechies (dbN2) mother functions. Therefore, a total of 20 features for each IE signal were extracted for each IE signal [34].

To compare the performance of DLMs to existing machine learning methods, features proposed in [33,34] were extracted from IE dataset in this study. A set of classifiers were generated and tested in a similar manner to DLMs (training was done on features of seven specimens while the features extracted from the eighth one was used for testing). The results are shown in Table 6 and compared with the 1D CNN. Overall the proposed 1D CNN was superior in *all metrics* compared to machine learning models. As

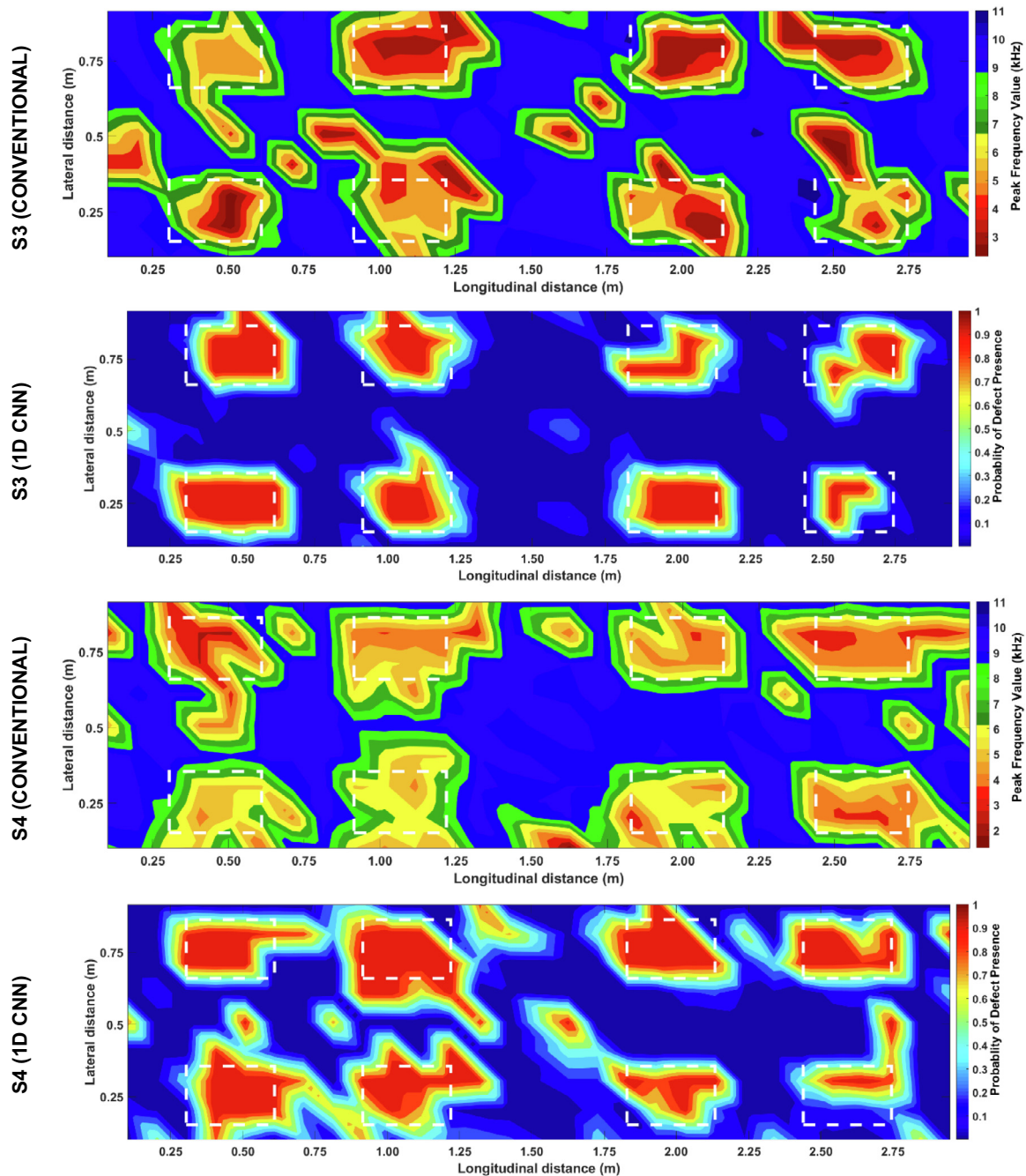


Fig. 12 (continued)

seen, the 2-feature classifier was unsatisfactory and ineffective for the IE dataset in this study; however, the 20-feature classifier was able to classify the IE data more consistently and produced comparable results to the proposed 1D CNN in this study. The comparison indicates the heavy reliance of conventional machine learning models to the feature extraction task in order to produce desirable results. Note that the process of choosing the classification features is fully automated in DLMs.

5.4. Analysis of a new IE dataset

The DLMs were successfully implemented for bridge evaluation in the previous section. However, the IE dataset was relatively humongous in terms of the defects (type and shape). Even though

there exist IE field dataset in some public repositories, such as FHWA InfoBridge [7], they are not validated with the ground truth. Therefore, the investigated DLMs could not be validated on such datasets. To mimic the field investigation, the models were used on a set of IE data acquired using a different IE device on a different bridge specimen. The bridge specimen was built in the Center for Transportation Infrastructure Systems at the University of Texas at El Paso, as shown in Fig. 13 [53]. The 0.22 m thick specimen was inspected by applying an IE device on 126 grid points with a sampling frequency of 400 kHz for 5 ms of data acquisition.

First, the IE data of the specimen were analyzed using the peak frequency analysis which resulted in a defect map shown in Fig. 14a. The user tailored frequency method detected most of the defected areas. The proposed 1D CNN, trained on all IE data

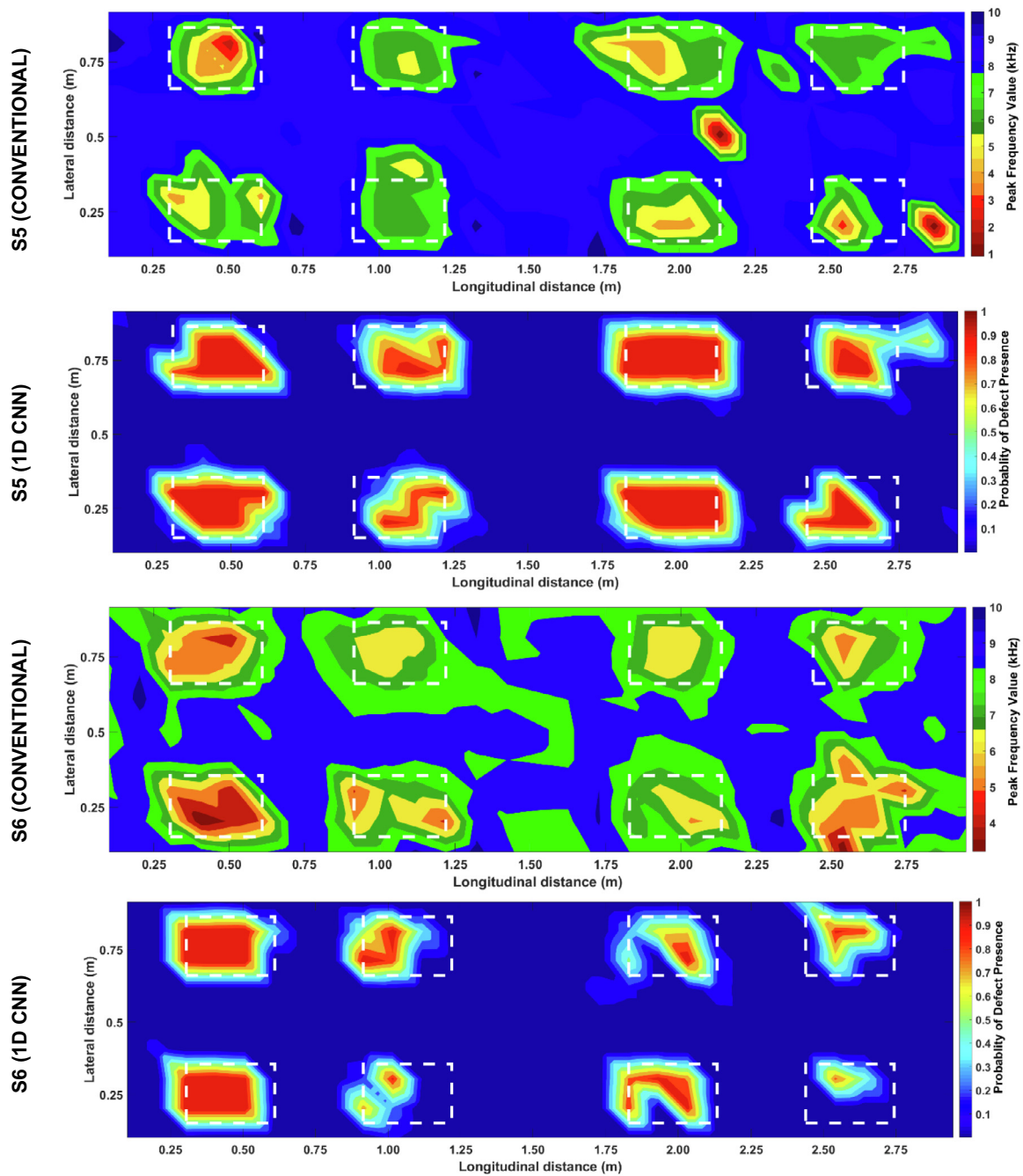


Fig. 12 (continued)

in this study, was directly applied on the IE data of this specimen to generate defect maps based on the probability of defect presence as show in Fig. 14b. The 1D CNN detected most of the defected area of the specimen; however, it reported several false positives. With a larger and more diverse annotated dataset it is likely to acquire substantially more accurate defect maps. The machine learning model with 20-feature classifier in Section 5.3.2 was also tested on this dataset which labeled all signals as defected concrete (Class 1). Note the 20-feature classifier produced comparable performance metrics as the 1D CNN when used on the IE laboratory data but yet failed to make a single true positive when used on the new dataset.

6. Conclusion

The feasibility of using deep learning models (DLMs) to classify impact echo (IE) data on bridge decks to localize subsurface defects has been investigated in this paper for the first time. Laboratory-made concrete specimens with artificial defects were inspected using IE. One-dimensional and 2-dimensiona DLMs were developed and applied to IE signals and their spectrograms, respectively. The proposed 1D CNN outperformed other investigated models in all performance metrics while required significantly less time to be trained. It was also superior to the conventional machine learning models used for IE classifications in the past. The defect maps

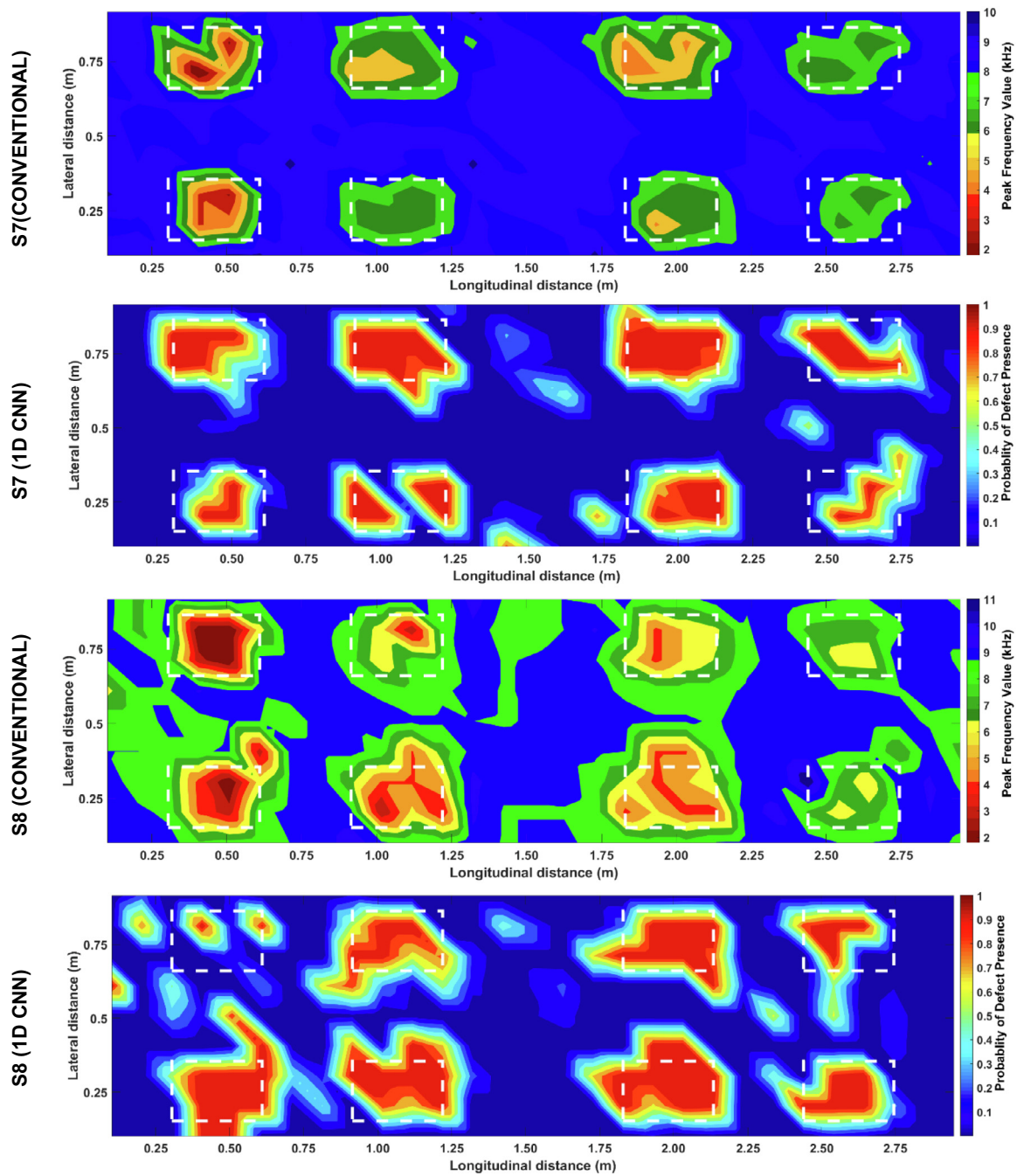


Fig. 12 (continued)

Table 6

Performance metric comparison between supervised and unsupervised learning models.

Model	Metrics															
	TPR				TNR				ACC				F1			
	Mean	COV	Max	Min	Mean	COV	Max	Min	Mean	COV	Max	Min	Mean	COV	Max	Min
1D CNN	0.70	0.14	0.81	0.48	0.95	0.04	1.0	0.85	0.88	0.04	0.92	0.83	0.75	0.08	0.83	0.65
2-feature [33]	0.12	0.98	0.34	0.01	0.94	0.03	0.98	0.88	0.73	0.05	0.79	0.69	0.16	1.02	0.45	0.03
20-feature [34]	0.68	0.07	0.73	0.61	0.93	0.04	0.98	0.87	0.87	0.02	0.89	0.83	0.73	0.04	0.77	0.69

generated by the 1D CNN matched or exceeded the ones generated by the peak frequency method which is the conventional method of IE analysis. To mimic field inspections, the proposed 1D CNN was used to classify IE data from a validation specimen that had

a different geometry (both in specimen and defects), and data acquisition device. The 1D CNN detected most of the defected areas although it reported more false positives than the conventional peak frequency method. This study showed the capabilities of deep

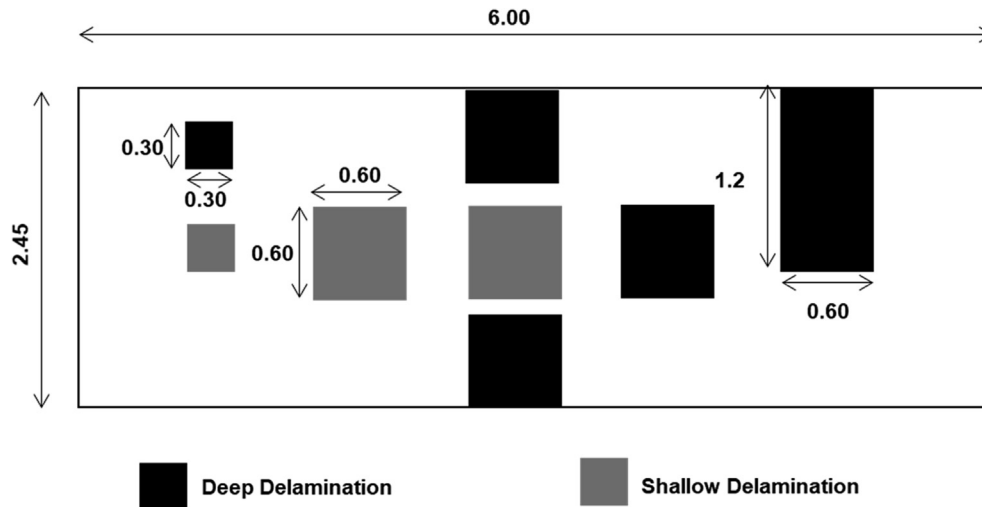


Fig. 13. Schematic plan of the validation bridge specimen with delamination.

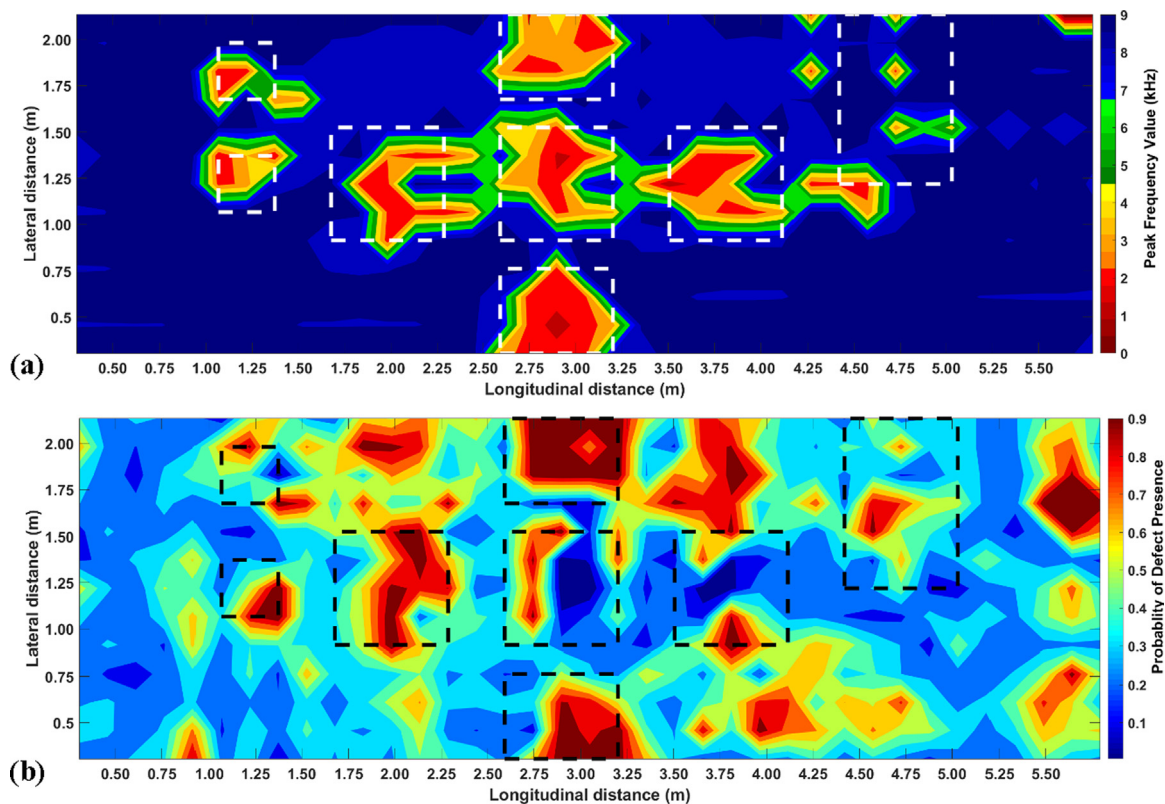


Fig. 14. Defect maps of the validation specimen using (a) frequency method, (b) 1D CNN.

learning for evaluation of bridge decks using IE; however, DLMS should be developed based on other non-destructive evaluation methods for more automated and robust evaluations. The deep learning models become more accurate as their training datasets grow larger. Therefore, forming accurately annotated datasets, particularly from the field, should be considered in the future.

CRediT authorship contribution statement

Sattar Dorafshan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Hoda Azari:** Conceptual-

ization, Resources, Validation, Supervision, Project administration, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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